experience in changing the orbits of various asteroids and comets of different compositions and strengths. We try to determine which ones can be pushed around and which cannot. By the twenty-second century, perhaps, we move small worlds around the Solar System, using (see next chapter) not nuclear explosions but nuclear fusion engines or their equivalents. We insert small asteroids made of precious and industrial metals into Earth orbit. Gradually we develop a defensive technology to deflect a large asteroid or comet that might in the foreseeable future hit the Earth, while, with meticulous care, we build layers of safeguards against misuse.

Since the danger of misusing deflection technology seems so much greater than the danger of an imminent impact, we can afford to wait, take precautions, rebuild political institutions—for decades certainly, probably centuries. If we play our cards right and are not unlucky, we can pace what we do up there by what progress we're making down here. The two are in any case deeply connected.

The asteroid hazard forces our hand. Eventually, we must establish a formidable human presence throughout the inner Solar System. On an issue of this importance I do not think we will be content with purely robotic means of mitigation. To do so safely we must make changes in our political and international systems. While much about our future is cloudy, this conclusion seems a little more robust, and independent of the vagaries of human institutions.

In the long term, even if we were not the descendants of professional wanderers, even if we were not inspired by exploratory passions, some of us would still have to leave the Earth—simply to ensure the survival of all of us. And once we're out there, we'll need bases, infrastructures. It would not be very long before some of us were living in artificial habitats and on other worlds. This is the first of two missing arguments, omitted in our discussion of missions to Mars, for a permanent human presence in space.

OTHER PLANETARY SYSTEMS must face their own impact hazards—because small primordial worlds, of which asteroids and comets are remnants, are the stuff out of which planets form there as well. After